1. Pre-analysis steps

After collecting the raw survey data, we performed an initial cleaning process to ensure that all responses met the criteria for inclusion in our study. Specifically, we removed three rows corresponding to respondents who indicated that they were not full-time undergraduate students at the University of Virginia, going from 79 rows of data to 76. This ensured that our analysis focused exclusively on the target population: full-time UVA undergraduates.

2. Analysis methods

We plan to use several methods to explore patterns and test our hypothesis. First, we will conduct a t-test to determine whether the observed proportion of students who answer “Yes” differs significantly from our hypothesized value of less than 50%. This will allow us to formally test whether fewer than half of students consider a hot dog to be a sandwich.

Next, we will create circle plots segmented by class year to visualize the proportional distribution of “Yes” and “No” responses within each group. These plots will help us compare trends across different years.

Finally, we will construct bar charts using raw counts to display the total number of responses for each category. These charts will provide a clear overview of how opinions are distributed across class years.

We also plan to run a Spearman analysis along with a chi-square analysis to see if there is any statistically significant correlation between a UVa student's year and whether or not there is any correlation between year and belief that a hot dog is a sandwich.

3. Evaluation of Success

To evaluate the success of our analysis, we will examine both the statistical results and the visualizations. We will use a significance threshold of $p = 0.05$ to determine whether the observed proportion of “Yes” responses is statistically different from 50%.

If the p-value is less than 0.05, we will interpret this as statistically significant evidence that the proportion of students who consider a hot dog to be a sandwich is less than 50%, supporting our hypothesis. If the p-value is greater than 0.05, we will conclude that there is insufficient evidence to support our hypothesis.

Additionally, we will assess whether the visualizations clearly display trends or patterns across class years. If no clear patterns are visible, we will avoid drawing strong conclusions from the data.

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Number of UVA students (n): 76
Number saying 'Yes': 18
Sample proportion p̂: 0.2368
z statistic: -5.3962
One-sided p-value (p < 0.5): 3.40371e-08
Two-sided p-value: 6.80741e-08
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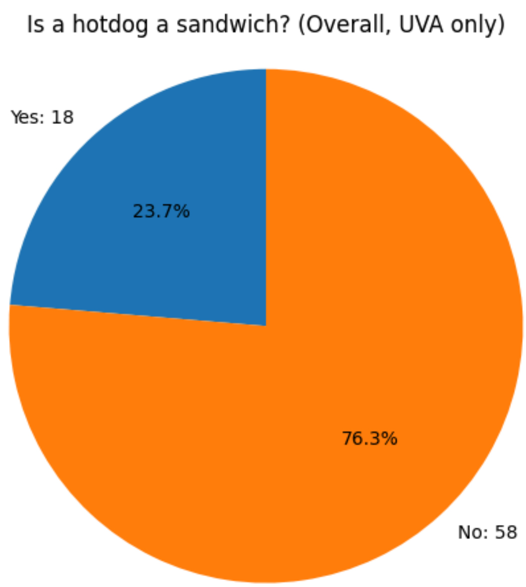
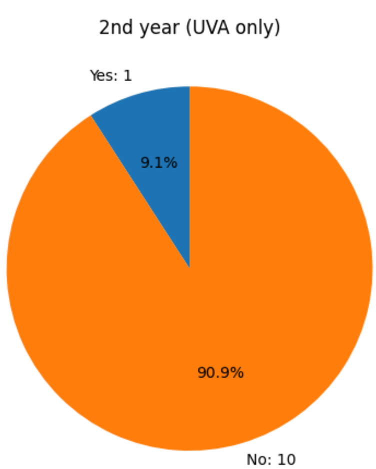
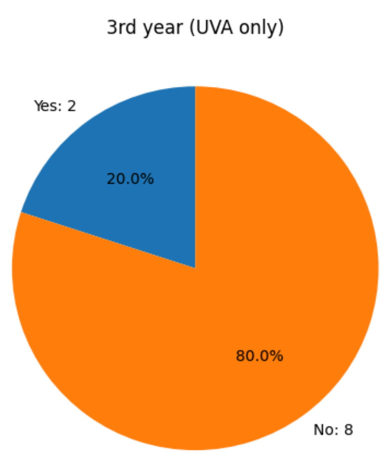
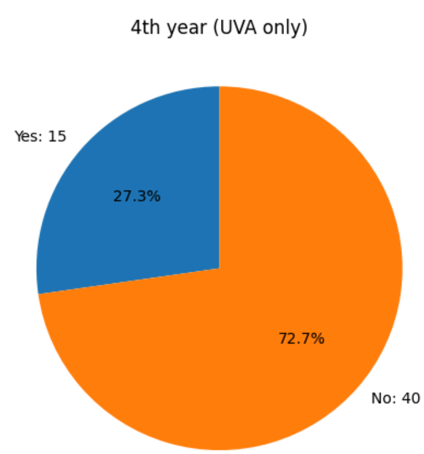
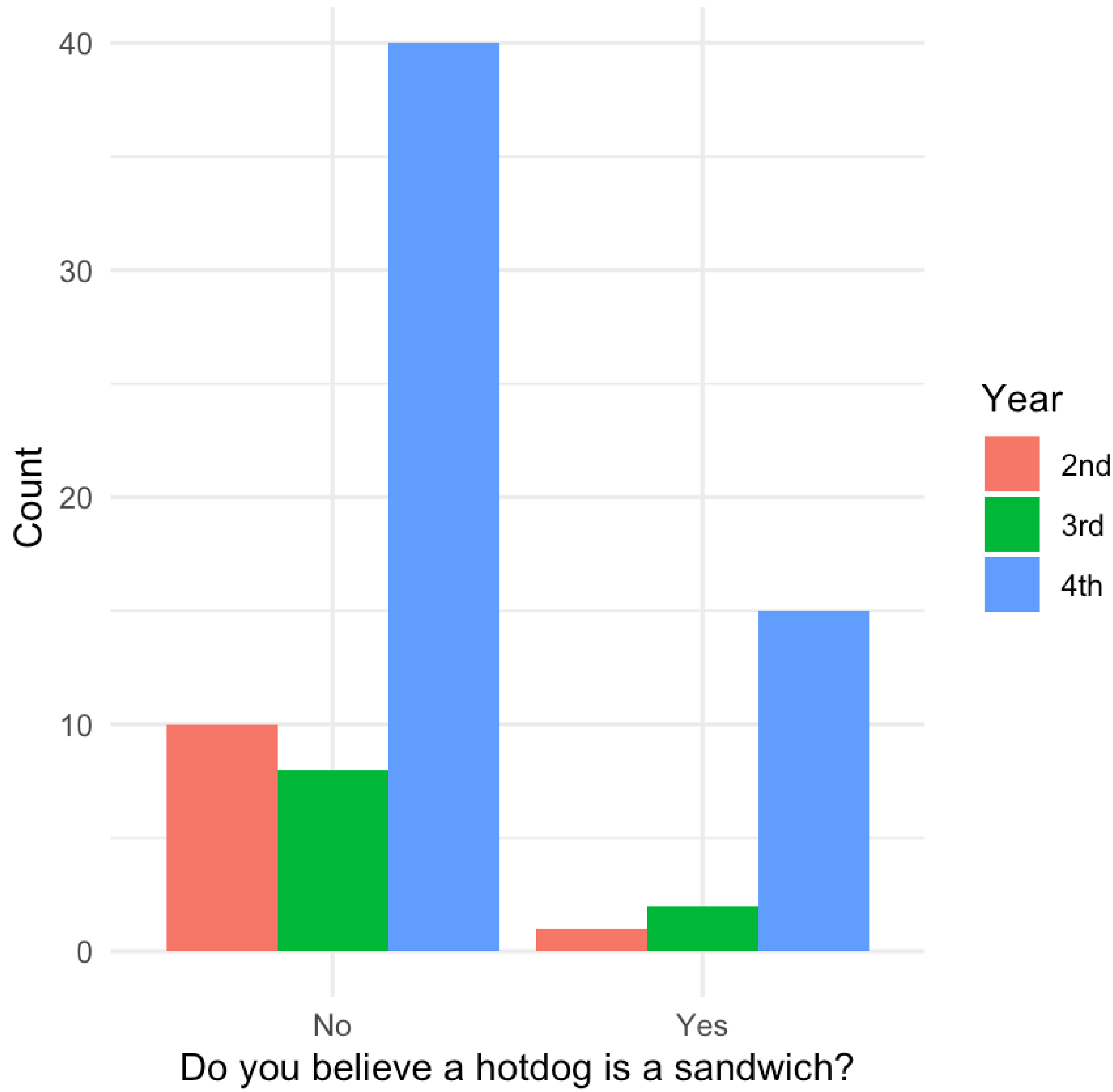








*Note: there were no 1st year responses to the survey

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Spearman Rank Correlation Results (UVA Only)
Rho: 0.146
P-value: 0.2094

Chi-Square Test Results (UVA Only)
Chi-square statistic: 1.763
P-value: 0.4142
Degrees of freedom: 2
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Spearman and Chi-Square Analysis Results

Neither the Spearman correlation nor the chi-square test provides evidence of a statistically significant association between year in school and belief that a hot dog is a sandwich.

T-Test Results:

A one-sample t-test was conducted to determine whether fewer than half of full-time UVA undergraduate students believe that a hot dog is a sandwich. After removing non-UVA undergraduate responses, the final sample consisted of 76 students. Responses were coded as binary values, with “Yes” = 1 and “No” = 0, resulting in a sample mean proportion of 0.237. The null hypothesis tested whether the true proportion was equal to 0.5, while the alternative hypothesis stated that it was less than 0.5. The test produced a Z-statistic of –5.39 and a p-value of approximately 3.4×10^{-8}. Since this p-value is far below the 0.05 significance threshold, we reject the null hypothesis and conclude that significantly fewer than half of full-time UVA undergraduates believe that a hot dog is a sandwich.